Task table

Task No.	Task name	Main objective	Respective experiments
Task 1	Simulation Environment and Ray-Tracing Infrastructure	Configure Sionna RT, validate the computational backend and establish the propagation simulation environment.	01, 02, 03
Task 2	Propagation Path Extraction and Characterisation	Extract, analyse and classify LOS and NLOS propagation paths and investigate their interactions with objects in the environment.	04, 05, 07, 08, 09
Task 3	Channel Impulse Response Modelling	Develop and validate the channel impulse response using sampled and fractional-delay representations.	06, 10, 11
Task 4	OFDM Channel Estimation and Frequency-Domain Validation	Implement OFDM channel estimation and validate the relationship between frequency-domain and delay-domain channel representations.	12, 13, 14
Task 5	Noise and SNR Evaluation	Evaluate multipath detection reliability, false detections and delay estimation accuracy under different noise conditions.	15, 16, 17, 18
Task 6	Target Detection and Range Analysis	Investigate sensing geometry, delay-to-range conversion, target detection and limitations of multipath-based localisation.	19, 20, 21, 22
Task 7	SDR-Based Experimental Validation	Validate the simulation framework using two USRP B210 devices, including RF transmission, reception, OFDM processing and hardware comparison.	23, 24, 25, 26, 27